TourCount 3.7.3

1. Introduction

TourCount should support you when recording European butterflies and day-flying moth (**Fig. 1**). It allows to register individuals species-specific, separated by sexes and metamorphic stages as well as individually with state and location in the field. It offers prepared data outputs and can substitute your field book and pencil, and if applicable a camera for documentary pictures.

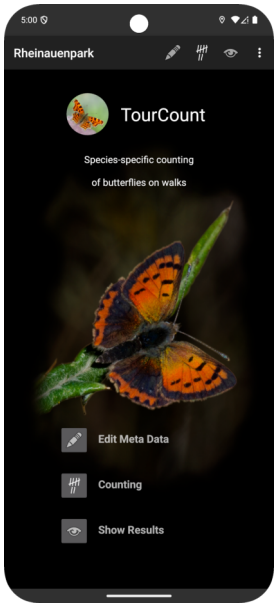

Fig. 1: Starting Page

The integrated database is tour-related, thus pro tour a new instance of a prepared database will be used. Databases can be individually created and adapted regarding expected butterfly species. The recorded data (metadata, counts and remarks) may either be read from the results page of the app or transferred to a PC for your own processing. The results page is prepared for an easy manual transfer into a butterfly registration system (e.g. www.science4you.org).

The app is published on https://github.com/wistein/TourCount with source code and documentation. It is open source, has neither tracking nor advertising functions and does not make use of Google Play Services. It demands only for permits which are needed for recording and managing the data:

- Access rights for storage,
- location service with GPS,
- optional internet access for reverse geocoding (address data derived from GPS coordinates by Nominatim service of OpenStreetMap.)
- Wakelock to prevent the app from self dimming or turning off, but turns off the screen and its touch sensitivity when held close to the body or in a jacket pocket until the device is used again.

2. Setting up

For installation hints refer to **chapter 5**.

When starting the app for the first time, you must grant permissions for storage and precise location tracking.

Before initial use you should adapt the app settings (→ **4. Further Functions**).

The preconfigured species list should be adapted to the species to be expected regionally. For this use the editing icons in the app bar of the counting page (**Fig. 2**):

- ⊕ "**Add Species**",
- 🗑 "**Remove Species**" or
- ✎ "**Edit Terms**"

On these 3 pages the selection of species can be eased with **preselection** by 2 initial letters of the genus name.

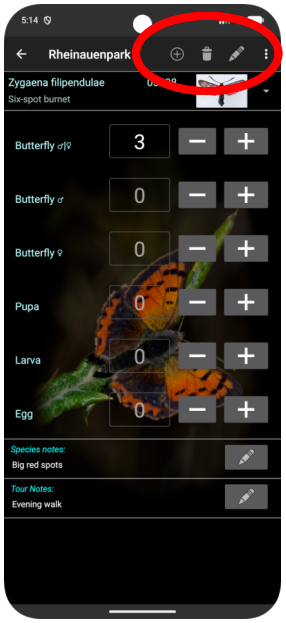

Fig. 2: Counting Page

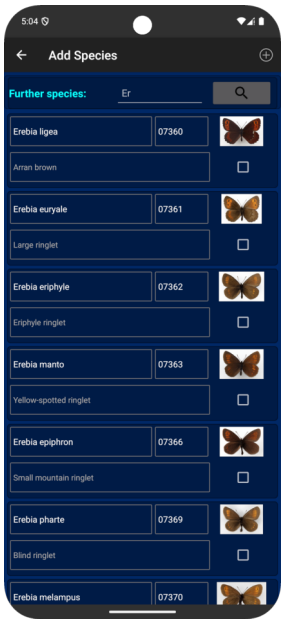

Fig. 3: "Add Species" page

Under "**Add Species**" (**Fig. 3**), further species can be added from the larger integrated list of European species using the checkboxes. (The added species then disappear from the selection list).

A placeholder for an indeterminate species (NN) can be added from the end of the "**Add Species**" scroll-down list, which can be edited under "**Edit terms**" once it has been defined.

Under "**Remove Species**" mark all irrelevant species and remove them with the 🗑 button.

Under “**Edit Terms**” if necessary, adjust tour name, comment, species name and code. (Species names, scientific and English and species code, 5 digits, with leading zeros).

. . .

. . .

Fig. 4: Species list (partly)

The unique codes will be used as a reference to show corresponding butterfly pictures on the counting and results page and as an option to sort the list. The codes derive from the numbering scheme of European butterflies by Karsholt/Razowski, as used e.g. in the German Lepiforum (https://lepiforum.org/).

The *-sign attached to code 07001 in **Fig. 4** indicates a complex of species that are difficult to distinguish. For sorting purposes, the largest code within such species complexes should be used for the group.

Caution: An incorrect code will display an “N/A” or incorrect image. If necessary, see the “**List of coded Butterflies.pdf**” on https://github.com/wistein/TourCount/tree/master/docs.

Changes are applied using the corresponding ⊕, 🗑 or 💾 function buttons.

Alternatively, a more comprehensive or specialized sample **species list** can be downloaded from https://github.com/wistein/TourCount/tree/master/docs, copied to the app's data directory “**Documents/TourCount**” and imported from there, and then customized. (This directory is created when the app is first launched. It is not deleted when TourCount is uninstalled; the data stored there is retained). Copy it to the app’s data directory **Documents/TourCount** and import and edit it in TourCount suitably. Changes to species always affect all existing section lists.

Then you may enter the observer name on the “**Edit Metadata**” page (**Fig. 5**) and finish this by clicking the saving symbol. If **reverse geocoding** is activated under “**Settings**” location-related metadata will be derived from GPS coordinates and will be inserted automatically. The editable metadata may also be modified anytime later.

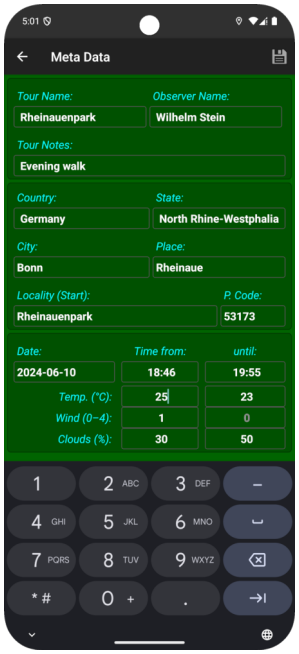

Fig. 5: Edit Metadata page

When you have created the counting list for all expected species and entered the basic metadata, the internal database is ready for export as a loadable "**Basic Database**" by the function "**Export as Basic DB**" in the menu of the starting page (**Fig. 9**). This saves a copy of the prepared empty database as **tourcount0_dl_Tourname.db** within the app’s files directory "**Documents/TourCount**".

The basic database does not contain any count data, comments, or inspection-specific metadata. It just stores the structure of the set up DB and serves as a template for further tours recordings in the region. It should be renamed suitably, e.g. tourcount0_dl_region xyz.db.
Note: The file name of a **Basic DB** must always start with the string “**tourcount0**“.

The data directory “**Documents/TourCount”** is created during the first app call. When uninstalling TourCount you will not loose your data as this directory remains untouched.

The naming convention for filenames of the exported TourCount files:

- Basic DB: **tourcount0_dl_Tourname.db**
- Current DB: **tourcount_dl_Tourname_StartingDate.db**
- Result: **Tour_dl_Tourname_StartingDate.csv**
- Species list: **species_dl_Tour_Tourname_currentDate.csv**

With "**dl**” = data language of local species names (de, en, fr, it, es) and blue = fixed strings.

3. Usage

After starting the app, a message about the first GPS fix is displayed and, if applicable, with a message indicating that the detected position is outside the European fauna area (as a reminder that only European species are supported).

Start with "**Edit Metadata**" (**Fig. 5**). Enter the specific metadata of the tour and save them.

Tip: Date and time can be entered by tapping the relevant field. If a different date or time is to be entered, these fields can be pressed longer for an input dialog.

Continue with "**Counting**" (**Fig. 2**). Select the species by clicking the species line to scroll down the species list.

To count just tip on the (**+**)-**button** of the corresponding category (♂|♀, ♂, ♀, pupa, larva, egg) of the species. The counter will increment and a page opens to enter the **individual data** (**Fig. 6**). Location info, latitude, longitude, height as well as date and time will be added automatically. Location may be edited and condition rating (state 0-6 with 0 = undefined, 1 = very good) as well as multiple counts may be applied here.

To accept the count go back to the counting page by clicking the 🖫 **button**.

The (**-**)-**buttons** on the Counting page allow for corrections. Mind that the (**-**)-button reduces the individuals list in a last-in-first-out-mode for the respective category.

If you enter a "0" in the "Remarks" field for the species on the counting page, the species will also be included in the results later.

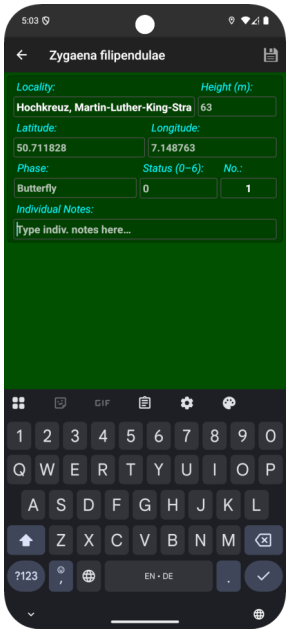

Fig. 6: Edit Individual Data page

To move back one page you can use the back-button or the arrow in the left upper corner. To make sure to save edited content you should go back by tipping the 🖫 or 🗑 **button**.

Before finishing TourCount after a tour you should complete the metadata and save the current reckonings by "**Export Results**" from the menu of the starting page (⇨ **Tour_dl_Tourname_YYYYMMDD_hhmm.csv** in the directory **Documents/TourCount/**). If no start time is specified, the current time will be used.

The "**Counting Results"** page (**Fig. 7 and 8**) can be opened on the starting page with "**Show results**" or the 👁 **icon**. It shows all the registered data orderly arranged. Beneath the metadata it shows the totals followed by a list of all counted species with their individual records sorted by the recorded time.

With large lists or big amounts of data, the response of the app can be somewhat delayed, especially when calling up pages with large scroll lists, as extensive calculations are running in the background. If the message "TourCount not responding" appears, please reply with "**Wait**".

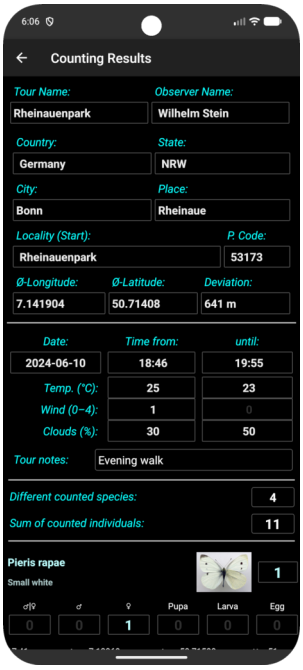

Fig. 7: Results page (head)

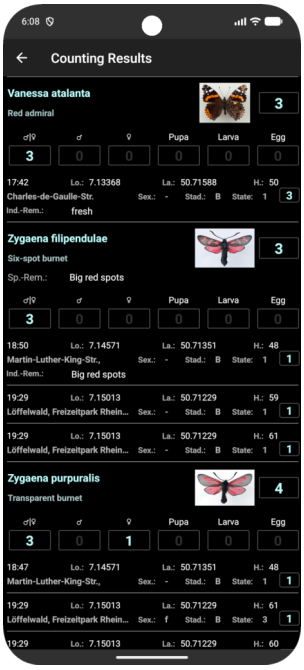

Fig. 8: Results page (continued)

4. Further Functions

The main menu on the starting page (**Fig. 9**) has Settings, Reset, Import, Export, Help and Info functions.

In "**Settings**" (**Fig. 10**) you may adapt the look an feel in some aspects to your wishes, e.g. sounds, sorting order, background or left-/right-hand counting.

"**Reverse Geocoding**"[1] allows for automatic insertion of statements of place (*postal code, city, place*) into metadata and *location* data into the individual's record.

For preparing a new tour you may use "**Reset Data**" to reset the tour-specific metadata and count data. Alternatively you may import a stored Basic DB, another tourcount-DB or a species list.

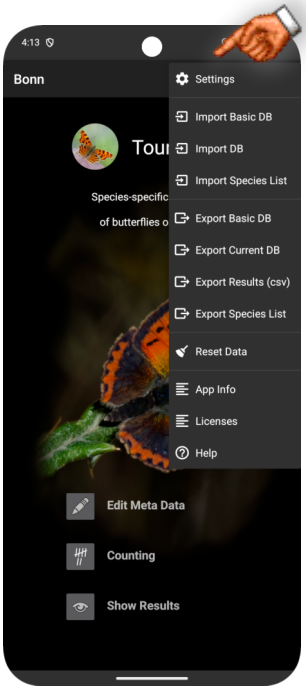

Fig. 9: Main Menu

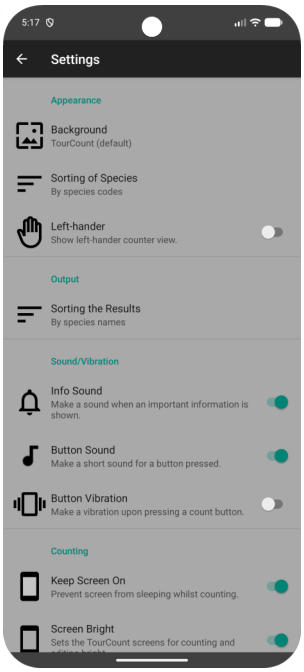

Fig. 10: Settings (partly)

Android-specifically, TourCount stores the data in the same, equally named SQLite DB file in the the app's own internal storage area. As this file cannot be read or changed directly by the user, importing or exporting the data from/to files in a user accessible storage area is necessary.

With "**Import Basic DB**" you can select and import an existing Basic DB file that was exported as a template for a regional tour and begins with "**tourcount0**".
Note: The file name must always begin with the exact character string "**tourcount0**" (e.g., "tourcount0_en_KottenForest.db"), otherwise the file cannot be imported as a Basic DB.

With "**Export Basic-DB**", the internal DB can be exported as the Basic-DB "**tourcount0_dl_Tourname.db**", i.e. without walk-specific data but with general metadata and the current species list to **Documents/TourCount**, e.g. if permanent changes have been made to the count list, such as adding new species (see "**2. Setting up**").

With "**Import DB**," an exported or appropriately named TourCount DB containing count data can be selected and loaded into the app.
Note: The file name must always begin with the exact character string "**tourcount_**" (e.g., "tourcount_en_KottenForest-20250815.db"), otherwise the file cannot be imported as a normal DB.

"**Export Current DB**" writes a copy of the entire DB with all currently contained data as "**tourcount_dl_Tourname_YYYYMMDD_hhmm.db**" into **Documents/TourCount**. The start date and the tour name get inserted if they exist in the metadata and, if necessary, the tourname cleaned up to remove characters that violate file naming conventions.

The function "**Import Species List**" lets you select and import an available species list. The metadata is retained during import, but all counting data is deleted. If the TransektCount app is installed, the exported species lists can be imported mutually.
Starting with app version 3.7.2, the species list has been updated to include the data language. The available data languages are German, English, French, Italian, and Spanish. When local species names were available, they have been included for the species listed in the app.
Note: The file name of a species list must always begin with "**species_**", otherwise the file cannot be imported.

[1] For Reverse Geocoding (to produce address info from GPS coordinates) the service of Nominatim from OpenStreetMap is used. A valid own email address is necessary for durably reliable queries of address data and to exclude abuse. The mail address will be treated confidentially and will only be used to contact you in case of service problems. For more info see https://wiki.openstreetmap.org/wiki/Nominatim.

With "**Export Species List**" the current species list without any other data can be exported as a CSV file (CSV = Comma Separated Values text file) "**species_dl_Tour_Tourname_YYYYMMDD_hhmmss.csv**". The tour name is also inserted here if it exists in the metadata, and any characters that violate the filename conventions are removed as necessary.
If you imported a species list that was exported from an app version earlier than 3.7.2, you can set its data language in Settings and then re-export the list. After that, the data language will be automatically detected and set in the future.

The function "**Export Results**" writes the metadata and the counting results into the preformatted spreadsheet-readable .csv-file "**Tour_dl_Tourname_YYYYMMDD _hhmm.csv**" to "**Documents/TourCount**". It tries to use the start date for the filename.
From here, the exported .csv result files can be viewed and, to a limited extent, edited in a spreadsheet app such as **Collabora** (open source, free, based on LibreOffice). It is available in the Collabora Office Store (an additional package source in the F-Droid app store), in the Play Store, or directly at https://www.collaboraoffice.com/downloads/).

The table can be sorted in advance using the result sorting option in TourCount (by species or codes), depending on your preference for easier data entry into a central monitoring web page like https://www.falterfunde.de/platform/s4y/falterfunde/index.do.

When loading the data into a spreadsheet, to ensure that the formats and character sets are displayed correctly, please ensure that

- file origin is "Unicode UTF-8",
- delimiter is comma and
- quotation marks ("") are set for text field recognition.

Fig. 11 shows a part of the results table imported into the Collabora app and formatted there.

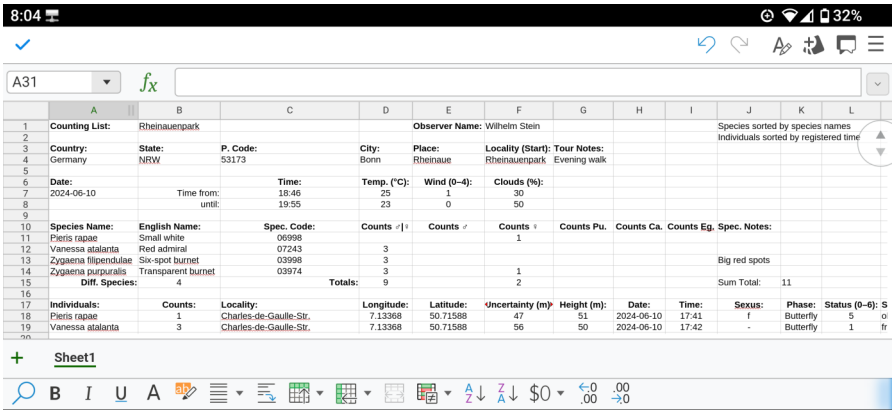

Fig. 11: In Collabora imported and formatted CSV table

If the leading zeros of the species codes or codes with * are to be displayed correctly, the option "Format fields in quotation marks as text" must be selected in the import dialog, if available, or else the column with the codes (in Fig. 11: Column C) must be formatted as text.

The exported .csv result files can also be transferred to a PC for further processing (see **6.1 Tips**) and importing into a spreadsheet program (tested with MS Excel, LibreOffice Calc, and Softmaker PlanMaker).

IT-affine users may also transfer the exported "**.db**" files to a PC and examine or edit them there with a free tool like "**DB Browser for SQLite**" (https://sqlitebrowser.org). It lets you edit them manually or per SQL script. Some useful example SQL scripts are published in the docs directory of the GitHub TourCount project site https://github.com/wistein/TourCount/tree/master/docs.

Under "**App Info**" you find the email address of the author and the history of the app development.

Under **"Licenses"** you find the license notes for code and figures.

The option menu on the counter page has a **photo function** to start the camera as quickly as possible without leaving TourCount.

A TourCount-specific message can also be sent there by **share function** via SMS or e-mail.

Optionally, the counter view is temporarily turned off by means of the **proximity sensor** when the phone is pocketed or closely held to the body. This saves energy, prevents accidental entries and brings the app back immediately when it is used normally again.

The app shows several **short pop-up messages** in 4 categories:

- **Error** (e.g. with a database).
- **Warning** (e.g. in the event of an invalid entry).
- **Note** (e.g. if an action might last longer).
- **Info** (e.g. after a successful process).

5. Installation hints

5.1 Install from F-Droid store (released version, recommended)

Get the TourCount app without docs or species list examples but with updates from the F-Droid app store:

https://f-droid.org/en/packages/com.wmstein.tourcount

Note on F-Droid:
Getting apps from F-Droid is at least as secure as getting them from the Google Play Store.
In contrast to the Play Store, all apps are also checked for privacy protection and compiled by F-Droid itself.
If an app does not meet all of F-Droid's requirements with regard to undesirable features, it is noted.

The source codes of the F-Droid apps are published and licensed with an open source license.

5.2 Get the app from the author's GitHub project pages (current development version)

App: https://github.com/wistein/TourCount/tree/master/apk

Download the **TourCount_xxx.apk** file from the GitHub page into the Downloads directory.
On the smartphone activate the Android option "Unknown source" in "Settings -> Security" before Installation. Then use the filemanager to install the app by clicking on the apk-file in the download folder.

Important: Finally deactivate the option "Unknown source"!

5.3 Setting up the app

After installation start TourCount once to create the necessary data directory **Documents/TourCount**.
Then get the documentation and relevant species lists from the author's GitHub pages.

Docs: https://github.com/wistein/TourCount/tree/master/docs

Copy these from the download folder into the new directory Documents/TourCount.
Now you can restart the app and finish setting it up.

5.4 Notes on updates

Unfortunately, the two installation methods mentioned above are not mutually compatible for updates, since although the .apk files are generated from identical source code, they are signed with different keys.

In the case of major version jumps with functional additions, structural changes may have been made in the internal database of an app. After such a change, the database version is incremented. This is recognized by the app and the currently used DB is adapted internally.

However, the currently adapted DB version cannot be used after a downgrade to a previous app which uses a previous DB version.

6 Annex

6.1 Tips

Transferring data between smartphone and PC

Connect the smartphone to the PC using a USB cable. In the smartphone notifications find „**Charging this device via USB**“, tab for more options and select „**Use USB for file transfer**“.

In Windows Explorer or under Linux in Nautilus, Nemo or similar file manager the smartphone is now shown with its technical ID. In the "**Internal shared memory**" area, the "**Documents/TourCount**" directory can be read and written to for data exchange.

6.2 Glossary

CSV file:
Comma-separated values file. Text-based file format for exchanging data in tabular form (e.g. for importing TourCount result data into spreadsheet programs).

File directories of TourCount:
The public app-specific files directory (for DB files and exported CSV files) is:

"Documents/TourCount”

Data stored here can also be read by other apps. The data is not automatically deleted when the app is uninstalled.

The previous app-specific directory for the DB files:

"**Android/data/com.wmstein.tourcount/files**"

is no longer used from TourCount version 3.4.5, as data stored here cannot be read by other apps in newer Android versions and was deleted when the app got uninstalled.

GitHub:
Is a file hosting site for software development projects including version control. It is free of charge for Open Source Projects. Eponymous was the version control and source code management system Git. Run by GitHub, Inc. from San Francisco, USA. Since December 2018 the company belongs to Microsoft. According to Microsoft GitHub will remain an independent platform.

Numbering scheme according to Karsholt/Razowski:
The entomologists O. Karsholt and J. Razowski developed a numbering scheme for all European butterfly species, that among others is used by the German Lepiforum. According to that numbering scheme TourCount uses Codes for the identification of species. However, this limits the utilization of TourCount to European faunal areas, as there is no adequate global scheme.

Open Source:
Source code of software, which can be edited and used publicly. Open source software can mostly be used free of charge and does not contain propriately licensed or closed source elements.

6.3 References

TourCount project:
The repository of the TourCount project is situated on https://github.com/wistein/TourCount.
It contains all published files with source code, configuration of the Android Studio Development platform, documentation and installable APK files.

TourCount Documents:
You find documents, example species lists, SQL scripts for manipulating TourCount DBs and other information under https://github.com/wistein/TourCount/tree/master/docs.

TransektCount
TransektCount is the complementary Android app to support transect counters in nature preserving projects according to the Butterfly Monitoring Scheme methodology in Europe. It allows a species-specific recording of individuals in all stages per transect section.

TransektCount project:
The repository of the TransektCount project is situated on https://github.com/wistein/TransektCount.
It contains all published files with source code, configuration of the Android Studio Development platform, documentation and installable APK files.

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